

Poisson Equation Solver

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1 Differential and Difference Form

The Poisson equation relates the electrostatic potential in a domain with the net charge in the simulation domain. To begin with let us start from the Gauss's law which states that

$$\frac{d}{dx}D(x) = \rho(x) \quad (1)$$

$$\frac{d}{dx}\varepsilon E(x) = \rho(x) \quad (2)$$

$$\frac{d}{dx}\varepsilon \frac{d}{dx}V(x) = -\rho(x) \quad (3)$$

where ρ is the volumetric charge density (L^{-3}), D is the electric flux and V is the electrostatic potential. Most of the times in the textbooks that above Poisson's equation is written as

$$\frac{d^2}{dx^2}V = -\frac{\rho}{\varepsilon} \quad (4)$$

where it is assumed that the permittivity is constant throughout the simulation domain. If such an equation is used we have to explicitly apply boundary conditions at the interface of the materials, which is not so convenient (more about this later).

So in order to solve the above equation, let us set a few notations. Consider a 1D simulation domain which is a line with discrete points with the spacing of Δx . Let the discrete points be labeled by i where i goes from 1 to n . The first and the last points are the boundaries of the simulation domain. Let us start with the eq. 3. Based on our discussion in the numerical differentiation we can write the LHS of the equation as at the point i as follows

$$\frac{d}{dx}\varepsilon \frac{d}{dx}V = \frac{\varepsilon_{i+0.5} \frac{d}{dx}V|_{i+0.5} - \frac{d}{dx}V|_{i-0.5}}{\Delta x} \quad (5)$$

$$= \left[\varepsilon_{i+0.5} \frac{V_{i+1} - V_i}{\Delta x} - \varepsilon_{i-0.5} \frac{V_i - V_{i-1}}{\Delta x} \right] \frac{1}{\Delta x} \quad (6)$$

In the above expression we require the value of the permittivity in the middle of the two points. For this we will simply assume that it is an average value of the two nearest neighbouring nodes. Thus now the above LHS expression is

$$\frac{d}{dx}\varepsilon \frac{d}{dx}V = \frac{1}{2\Delta x^2} [(\varepsilon_{i+1} + \varepsilon_i)V_{i+1} - (\varepsilon_{i+1} + 2\varepsilon_i + \varepsilon_{i-1})V_i + (\varepsilon_i + \varepsilon_{i-1})V_{i-1}] \quad (7)$$

The simplified form when the permittivity is constant

The right hand side is straight forward in the sense that the charge term is defined only at the node i . So the RHS is defined only at the particular i^{th} point

$$\rho(x) = \rho_i = e \left(N_{D,i}^+ - N_{A,i}^- + p_i - n_i \right) \quad (8)$$

where $N_{D,i}^+ - N_{A,i}^-$ is the net ionized doping at node i , p_i and n_i are the hole and electron concentration at the node i . So the complete Poisson's equation in the difference form is given by

$$\frac{1}{\Delta x^2} [(\varepsilon_{i+1} + \varepsilon_i)V_{i+1} - (\varepsilon_{i+1} + 2\varepsilon_i + \varepsilon_{i-1})V_i + (\varepsilon_i + \varepsilon_{i-1})V_{i-1}] = \rho_i \quad (9)$$

Please remember that the equation derived above is valid only for uniform meshing i.e. Δx is constant and for a particular mesh point i which is unlike the differential form.

2 Linear and Nonlinear forms

If the ρ_i is constant with respect to the potential then it results in the linear form of the Poisson's equation. This is the case of dielectrics with oxide charges etc. In such a case we can simply set up the system of equation of the form of $\bar{A}\bar{x} = \bar{b}$ and solve them where $\bar{A}$ is the discrete form of the derivative operator, $\bar{x}$ is the potential at the nodes and $\bar{b}$ is the right hand side, ρ at different nodes. The assembly of the system of equations and boundary conditions are discussed below.

In semiconductor devices, however, the ρ_i depends non-linearly(exponentially) on the electrostatic potential ($E_C = -qV$) thereby making the Poisson's equation a nonlinear differential equation. Hence to solve the discretized form of the Poisson's equation we need to use Newton-Raphson method (that was discussed earlier).

3 Boundary Conditions

To solve any differential equations (time independent) we need to apply boundary conditions. Generally these BC are applied at the end/edge of the simulation domain ($i = 1$ and $i = n$). However, if we do not consider that the permittivity is varying with the position then we also have to apply the BC at all the material interfaces. In this part we will discuss the BC.

3.1 At the end of Simulation domains

Whether the material is same through out the simulation domain or not we need to know some values of the electrostatic potential in the simulation domain.

3.1.1 Dirichlet Boundary Condition

The Dirichlet BC are the ones where we know the applied bias. For example, the the gate, drain, source and substrate contacts. Since at these contacts we are applying a physical battery or power supply we assume that we know the potential at these contacts/nodes. To understand how to apply the BC let us first of all recall that at equilibrium the Fermi level is constant through out the simulation domain. Another very important fact is that in physical systems the absolute energy levels are almost useless. However, what is important are the difference between the energy levels. Hence we need to first of all set a reference level with which we would measure everything. The most convenient according to me is the Fermi level in the device in equilibrium. Let us fix its value to 0 eV or Joules (**Please choose your units and stay with it. I would recommend to use SI units**). Another very important point that we will fix is that the electrostatic potential, V , is the negative conduction band edge, E_C . Of course you can set the V to E_i or E_v or even vacuum level but recall that all the values differ only by a constant which would get cancelled out when we take difference or calculate the derivative (which is nothing but a form of difference). There are some advantages of considering the $V = -eE_C$ which we will discuss when we study the Schrödinger's equation solution. So in equilibrium the boundary condition at the edges of the simulation domain would be nothing by the value of E_C with respect to E_F with a negative sign. Using simple Maxwell-Boltzmann statistics we can write it as

$$n = N_C e^{\left[-\frac{E_C - E_F}{k_B T}\right]} \implies -(E_C - E_F) = V_{BC} = \frac{k_B T}{e} \ln \left[\frac{n}{N_C} \right] \quad (10)$$

The factor of e is there for conversion from Joules to eV to Volts (negative sign). It is assumed that the contacts are in equilibrium hence the electron concentration is same as the doping i.e. N_D if the doping is donor type. If the doping is p-type then we can similarly calculate the value of the valence band edge, E_V , with respect to the E_F and hence calculate $E_C - E_F$

In the case of non-equilibrium, when we apply a bias we are lowering or raising the potential energy (depending on the applied bias) of the carrier. However, note that the carrier concentration remains the same and hence the difference between $E_C - E_F$ remains the same. This implies that the Fermi-level in the contact is also shifted by the same amount. Thus in the non-equilibrium case the boundary condition becomes

$$V_{BC} = V_{Applied} + [-(E_C - E_F)] = V_{Applied} + \left[\frac{k_B T}{e} \ln \left[\frac{N_D}{N_C} \right] \right] \quad (11)$$

Physical meaning: The potential energy gets added This again is the BC for the n-type contact and the above argument can be extended for the p-type contact as well using the logic discussed for the equilibrium case.

3.1.2 Neumann Boundary Condition

Another important boundary condition is called Neumann BC. This is applied when we really don't know the applied potential but we know that the electric flux, D .

$$\epsilon \frac{dV}{dx} \cdot \hat{n} = C \quad (12)$$

where $\hat{n}$ is the unit normal vector to the boundary surface. In most of the practical cases, the RHS is set to zero which results in the following condition

$$\frac{dV}{dx} \cdot \hat{n} = 0 \quad (13)$$

Physically this boundary condition means that we allow the E_C i.e. electrostatic potential, V , is allowed to float with respect to the E_F to maintain the charge neutrality in the source and drain regions. This implies that the electric field component normal to the interface is zero (or more generally the charge enclosed, Gauss's law). These can be thought of artificial boundary conditions and are applied to limit the simulation domain and perform the simulations. The example of BC where this can be applied are the non-gate edges of the gate oxide, edges of the source and drain regions where there is no source and drain contacts, edges of the diode. Of course choosing this BC is critical as it can affect the convergence and the simulation results. For example consider a real MOS Capacitor. The substrate is very thick which is there to give mechanical stability and nothing important related to the electronic transport occurs in that region. So in order to limit the simulation domain we can apply Neumann BC. If the substrate depth is not selected with enough depth then the electric field will not be zero and hence we will be forcing condition will be incorrect.

Another way BC that can be applied at the substrate contact is the Dirichlet BC. How ???

3.2 At the Material Interface

If we do not account for the permittivity material dependence in the Poisson equation then in such a case we have to explicitly introduce the BC at the material interfaces. From basics of electromagnetism we get the following (assuming that there is no charge at the interface)

$$D_{semi} \cdot \hat{n}|_{semi} = D_{ox} \cdot \hat{n}|_{ox} + Q_{int} \quad (14)$$

$$\varepsilon_{semi} \frac{dV}{dx} \cdot \hat{n}|_{semi} = \varepsilon_{ox} \frac{dV}{dx} \cdot \hat{n}|_{ox} + Q_{int} \quad (15)$$

$$(16)$$

Q_{int} is the interface charge.

4 Assembly and Application of Boundary Conditions

4.1 Linear Form

The discretized version of the Poisson equation at the i^{th} node can be written as

$$\frac{1}{2\Delta x^2} [(\varepsilon_{i+1} + \varepsilon_i)V_{i+1} - (\varepsilon_{i+1} + 2\varepsilon_i + \varepsilon_{i-1})V_i + (\varepsilon_i + \varepsilon_{i-1})V_{i-1}] = \rho_i \quad (17)$$

For the sake of simplicity let us rewrite the above equation as

$$t_{i,i+1}V_{i+1} - t_{i,i}V_i + t_{i,i-1}V_{i-1} = \rho_i \quad (18)$$

$$\begin{bmatrix} t_{i+1} & t_i & t_{i-1} \end{bmatrix} \begin{bmatrix} V_{i+1} \\ V_i \\ V_{i-1} \end{bmatrix} = [\rho_i] \quad (19)$$

Continuing this above for all the nodes except for the first and last node (which are the boundary nodes) we get the following matrix

$$\begin{bmatrix} \dots & & & & & & \\ t_{2,1} & t_{2,2} & t_{2,3} & & & & \\ \vdots & \ddots & & & & & \\ \dots & t_{i,i-1} & t_{i,i} & t_{i,i+1} & & & \\ & & \ddots & \vdots & & & \\ & & & t_{n-1,n-2} & t_{n-1,n-1} & V_{n-1,n} & \\ & & & & \dots & & \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ \vdots \\ V_i \\ \vdots \\ V_{n-1} \\ V_n \end{bmatrix} = \begin{bmatrix} ? \\ \rho_2 \\ \vdots \\ \rho_i \\ \vdots \\ \rho_{n-1} \\ ? \end{bmatrix} \quad (20)$$

How to apply the BC. We will simply concentrate on the Dirichlet BC. This means that the equation at the first node will be

$$1V_1 = V_{app,1} + [-(E_{C,1} - E_{F,1})] \quad (21)$$

At the last node will be

$$V_n = V_{app,n} + [-(E_{C,n} - E_{F,n})] \quad (22)$$

This implies that the diagonal elements of the LHS matrix would be unity for the boundary nodes.

The above system of equation can be written as

$$\mathbf{A}V = \rho \quad (23)$$

where A is square tridiagonal matrix with dimension of $n \times n$.

Three ways to assemble A Matrix

1. Two nested for loops
2. One for loop using the property of the matrix
3. Generic Way: Two for loops
 - Outer loop: Loop over the nodes
 - Inner loop: Loop over the connections.

$$\frac{1}{2\Delta x^2} [(\varepsilon_k + \varepsilon_i)V_k - (\varepsilon_k + \varepsilon_i + \varepsilon_i + \varepsilon_j)V_i + (\varepsilon_i + \varepsilon_j)V_j] = \rho_i \quad (24)$$

The above system of equations can be solved using any standard method like LU solver, Gauss Sidel method etc. These can be applied using the standard library LAPACK, or in Octave/MATLAB using backslash operator. **If time permits we might come to these topic again and discuss some of the methods to solve the system of equations**

4.2 Non Linear form

To solve the non-linear version of the Poisson's equation it is necessary to linearize it.

$$F_i \equiv \frac{1}{2\Delta x^2} [(\varepsilon_k + \varepsilon_i)V_k - (\varepsilon_k + \varepsilon_i + \varepsilon_i + \varepsilon_j)V_i + (\varepsilon_i + \varepsilon_j)V_j] - \rho_i(V_i) = 0 \quad (25)$$

We can linearize the above equation by assuming a small perturbation to the electrostatic potential ($V_i + \delta V$). Please convince yourself that linearization would result in the form that is given by the Jacobian.

$$\frac{\partial F_i}{\partial V_i} = -\frac{(\varepsilon_k + \varepsilon_i + \varepsilon_i + \varepsilon_j)}{2\Delta x^2} - \frac{\partial \rho_i(V_i)}{\partial V_i} \quad (26)$$

$$\frac{\partial F_i}{\partial V_j} = \frac{(\varepsilon_i + \varepsilon_j)}{2\Delta x^2} \quad (27)$$

5 Practical Notes